# CI load procurement: PyPSA components

WattTime Impact Metastudy

21 July 2025

# Disclaimer & Participants

## Disclaimer

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## Participants

(alphabetical order; **bold: participated**, otherwise excussed)

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- Pierre Christian
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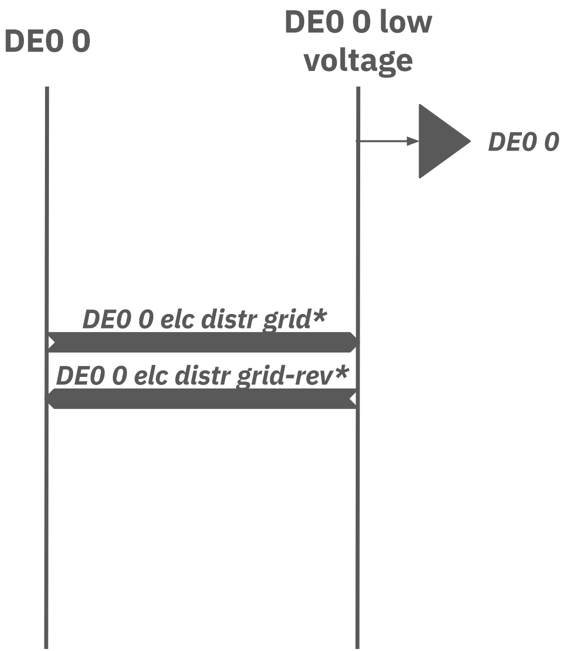
- Markus Groissböck
- Martha Frysztacki
- Virio Andreyana

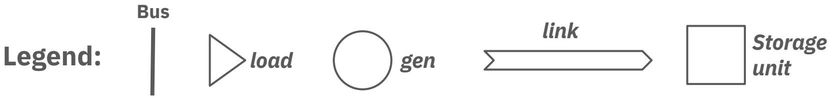
# CI loads

# Modeling of the CI loads

## 0-Traditional PyPSA-Eur loads



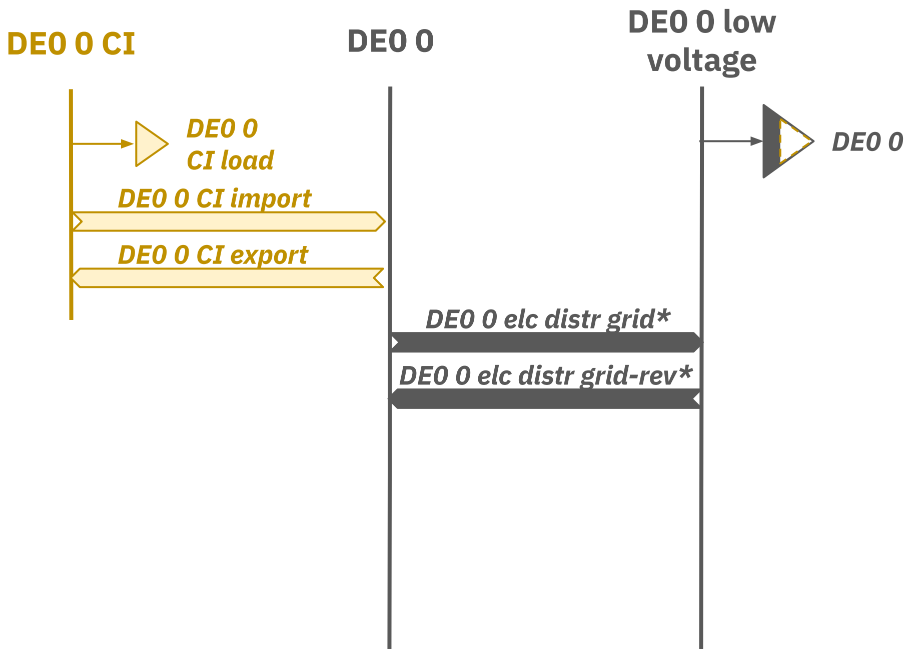

E.g., Germany CI (N.B.: only components that are additional to OS PyPSA-Eur are shown)




* electricity distribution grid (-reversed)

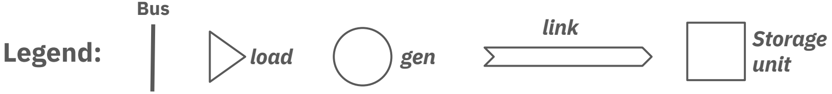
# Modeling of the CI loads

## 1-CI loads




**Step 1 - Add CI load**
- **CI load = fraction of the CI load** that is moved from low voltage **to** the **high voltage** side
- **Fraction** is retrieved **from dedicated data**
- **Profile** of the CI load is set in the **config**

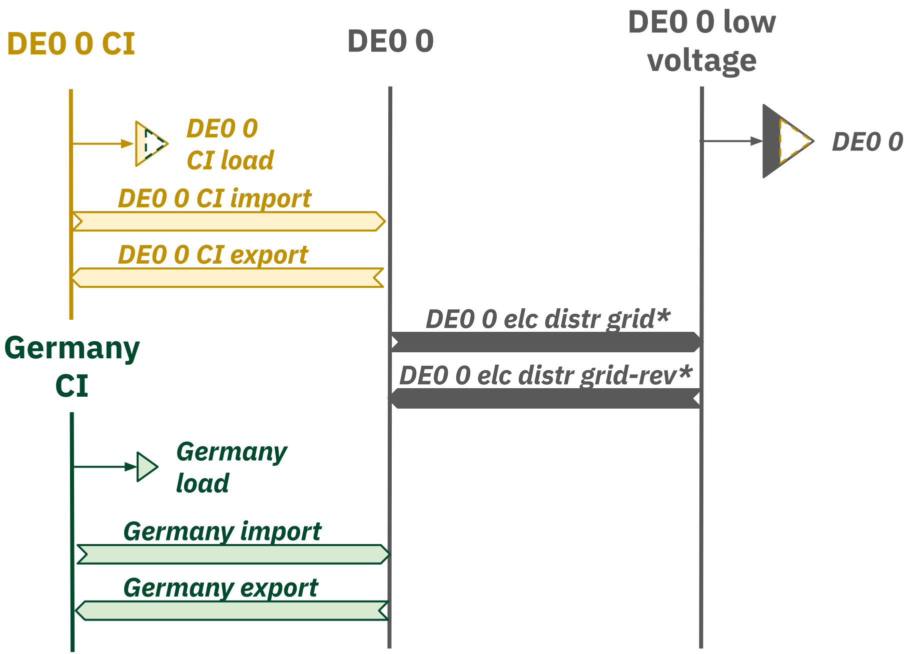
**E.g., Germany CI (N.B.: only components that are additional to OS PyPSA-Eur are shown)**




* electricity distribution grid (-reversed)

# Modeling of the CI loads

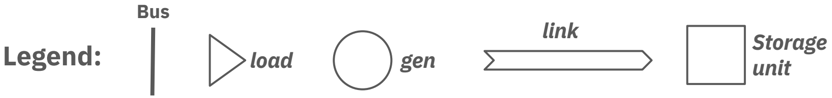
## 2-CI loads: participating to procurement




**Step 1 - Add CI load**
- **CI load = fraction of the CI load** that is moved from low voltage **to** the **high voltage** side
- **Fraction** is retrieved **from dedicated data**
- **Profile** of the CI load is set in the **config**

**Step 2 - Add participating CI**
- **Participating CI = fraction of the CI load** that participates to the procurement
- Participating **countries** are set in the **config**
- **Fraction** is set in the **config**

E.g., Germany CI (N.B.: only components that are additional to OS PyPSA-Eur are shown)

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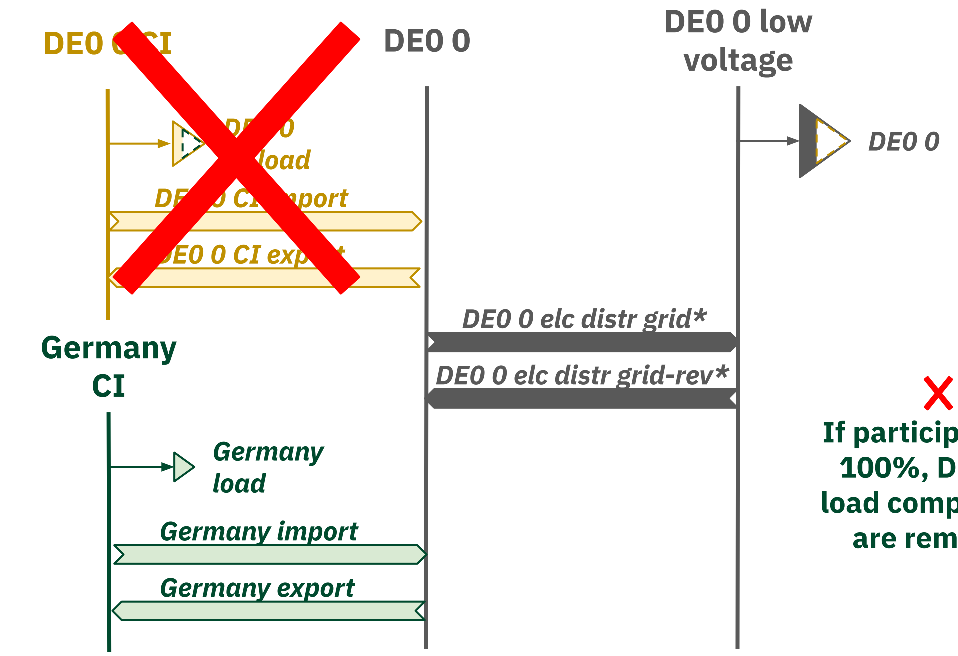
# Modeling of the CI loads

## 2-CI loads: participating to procurement (100%)




**If participation = 100%, DE 0 CI load components are removed**

**Step 1 - Add CI load**
- **CI load = fraction of the CI load** that is moved from low voltage **to** the **high voltage** side
- **Fraction** is retrieved **from dedicated data**
- **Profile** of the CI load is set in the **config**

**Step 2 - Add participating CI**
- **Participating CI = fraction of the CI load** that participates to the procurement
- Participating **countries** are set in the **config**
- **Fraction** is set in the **config**

**E.g., Germany CI (N.B.: only components that are additional to OS PyPSA-Eur are shown)**

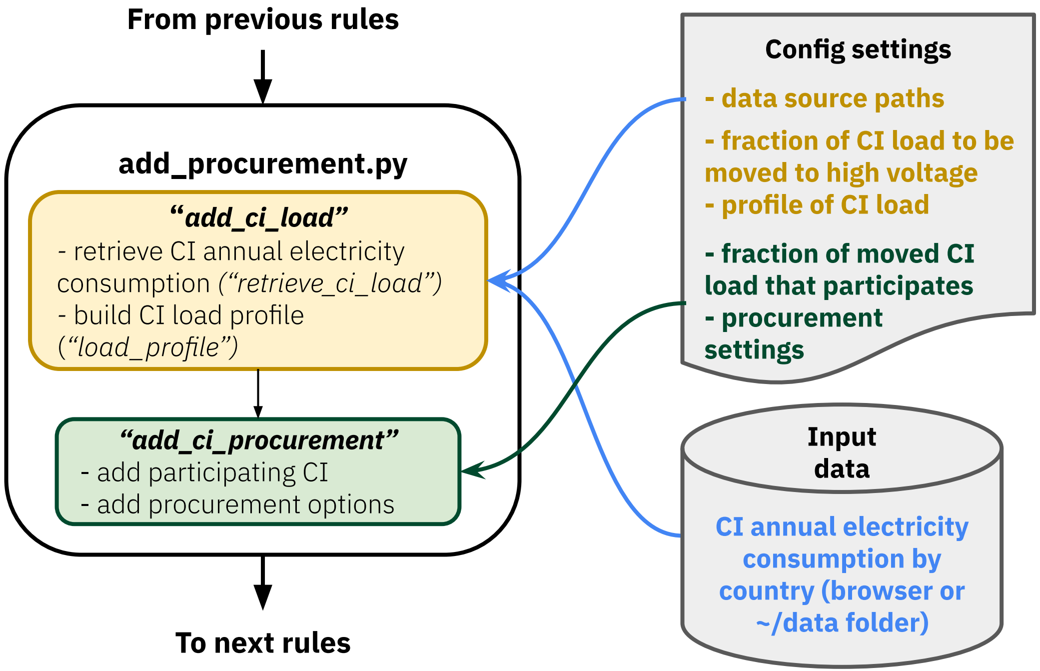


**Legend:**


*** electricity distribution grid (-reversed)**

# Modeling of the CI loads

## PyPSA-Eur workflow




### Step 1 - Add CI load

***“retrieve_ci_load”***:
**-in**: CI annual electricity consumption by country
**-out**: **share of CI consumption** over the total electricity by country

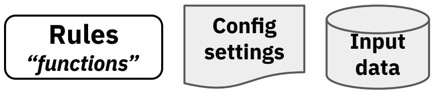
***“load_profile”***:
**-in**: share of CI consumption, CI settings
**-out**: **CI load time series (high voltage) by country**

### Step 2 - Add participating CI

**- add participating CI:** consider a **fraction of the CI** load from Step 1 in the countries **participating** to the strategies
**- add procurement options**: model the techs that can supply the participating CI loads



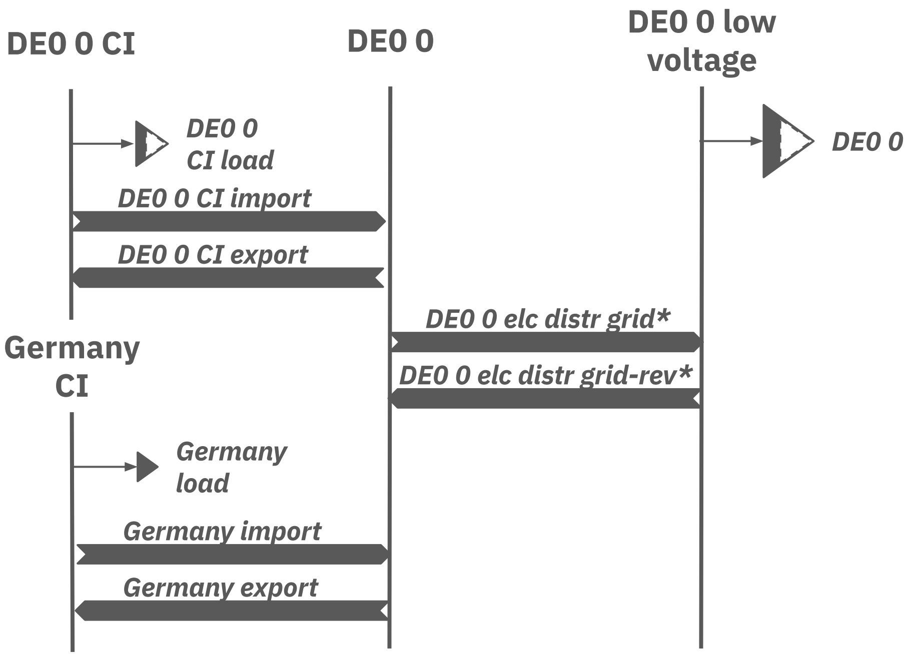
Legend:



# CI supply

# Modeling of the CI procurement supply

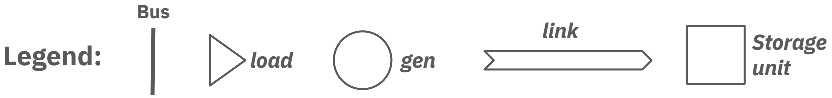
## Overview




**Step 3 - Add procurement options (part of "*add_ci_procurement*")**

- **Procurement options =** supply technologies involving
  - **generators**
  - **links**
  - **storage units**
- Spatial **scope:**
  - **node:** carbon free energy (CFE) can be only procured at the level of the participating CI load bus
  - **country:** CFE can be only procured at the level of the participating CI load bus
  - **all:** CFE can be only procured at the level of the participating CI load bus
  - **continent:** CFE can be procured anywhere in the system, by aggregating the procured energy as well as the participating CI load.

E.g., Germany CI (N.B.: only components that are additional to OS PyPSA-Eur are shown)

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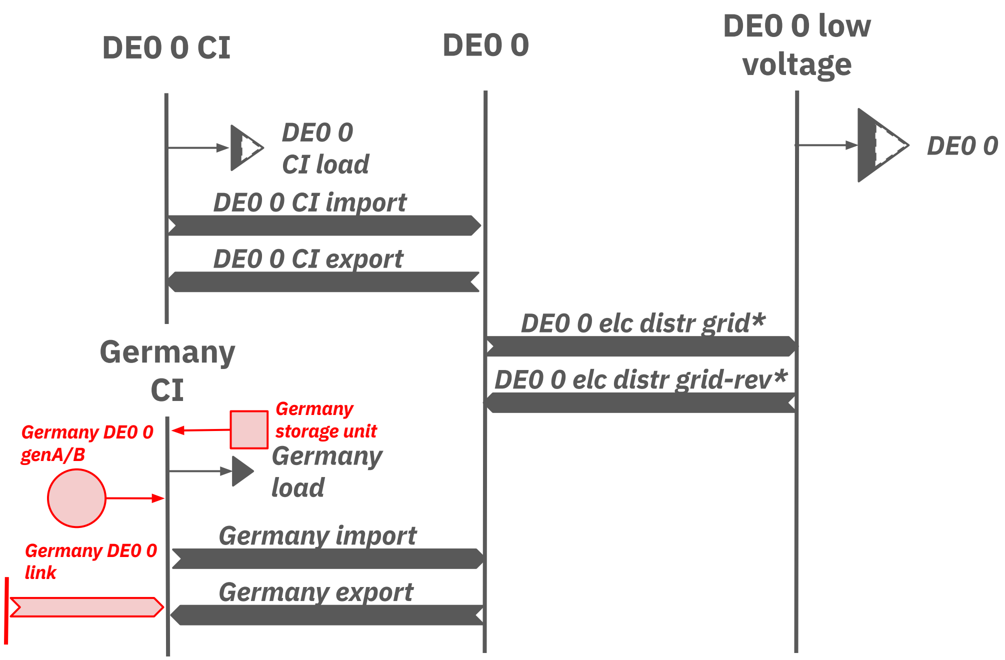
# Modeling of the CI procurement supply

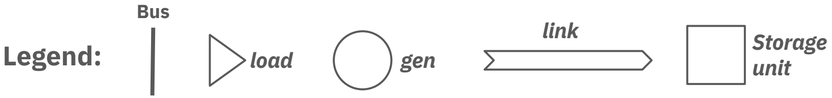
## Node




**Step 3 - Add procurement options (part of "*add_ci_procurement*")**

- **Procurement options =** supply technologies involving
  - **generators**
  - **links**
  - **storage units**
- **Spatial scope:**
  - **node**
  - **country**
  - **all**
  - **continent**
- **Options** are set in the **config.** E.g.,
  - **-genA = solar rooftop**
  - **-genB = onwind**
  - **-link = allam**

**E.g., Germany CI (N.B.: only components that are additional to OS PyPSA-Eur are shown)**

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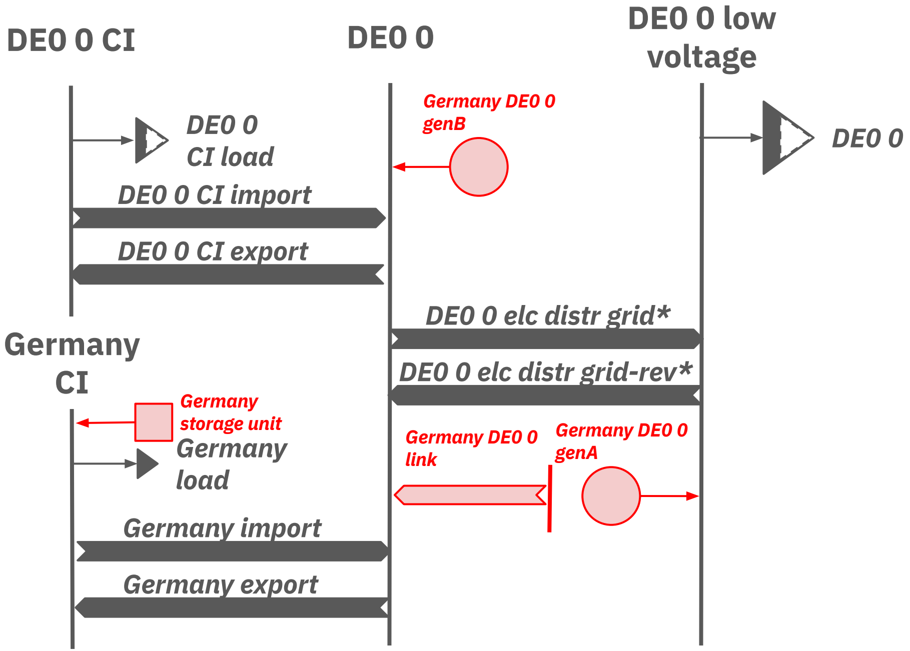
# Modeling of the CI procurement supply

Country

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**Legend:**


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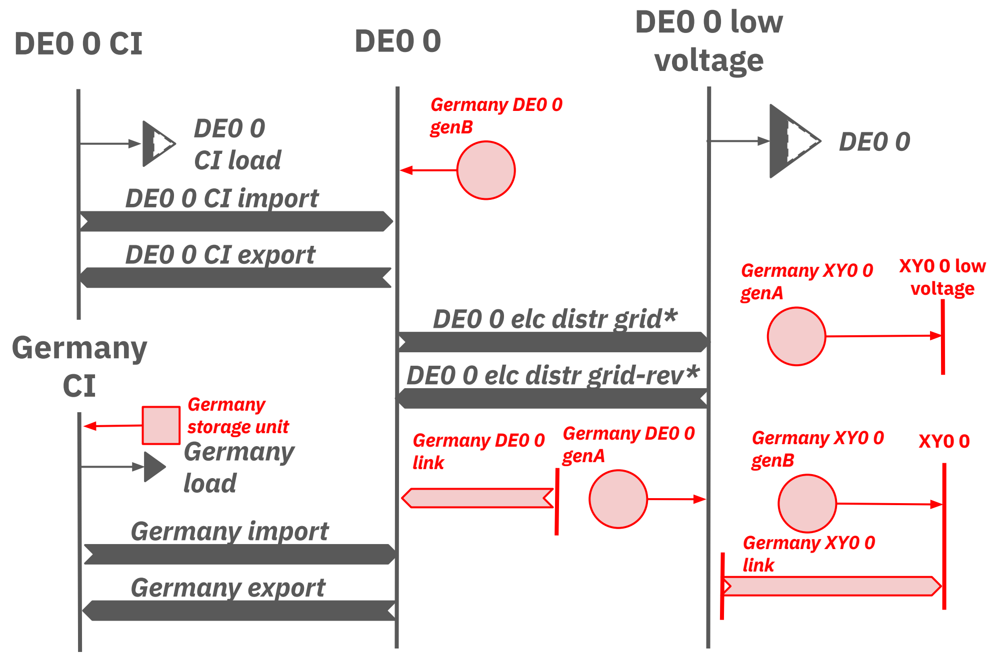
# Modeling of the CI procurement supply

All




**Step 3 - Add procurement options (part of "*add_ci_procurement*")**

- **Procurement options =** supply technologies involving
  - **generators**
  - **links**
  - **storage units**
- **Spatial scope:**
  - **node**
  - **country**
  - **all (N.B., XY = country other than DE)**
  - **continent**
- **Options** are set in the **config.** E.g.,
  - **genA = solar rooftop**
  - **genB = onwind**
  - **link = allam**

**E.g., Germany CI (N.B.: only components that are additional to OS PyPSA-Eur are shown)**



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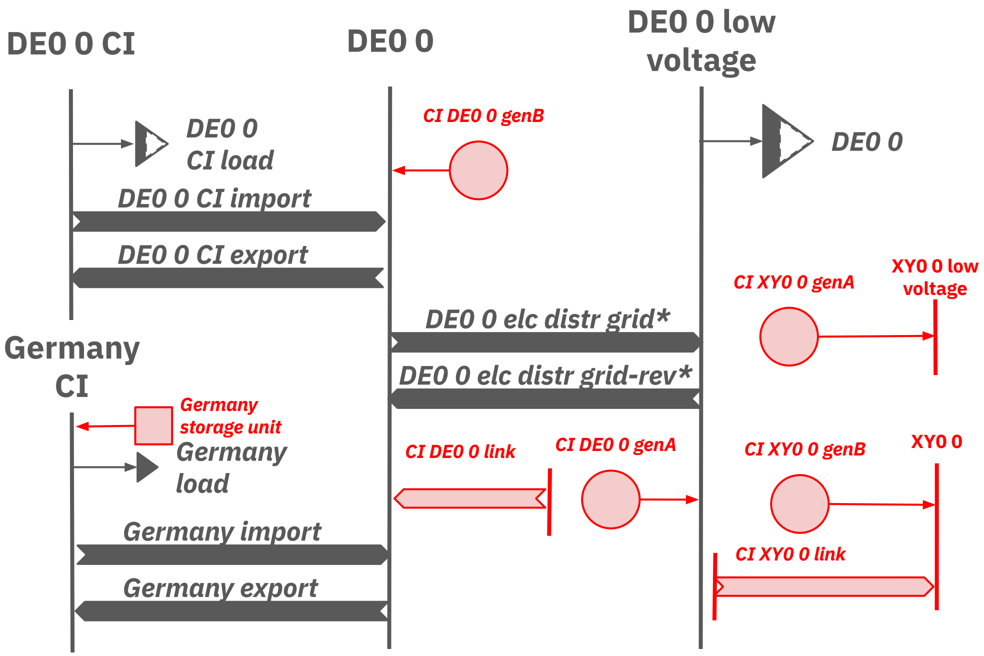
# Modeling of the CI procurement supply

Continent




E.g., Germany CI (N.B.: only components that are additional to OS PyPSA-Eur are shown)

**Step 3 - Add procurement options (part of "*add_ci_procurement*")**

- **Procurement options =** supply technologies involving
  - **- generators**
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  - **- storage units**
- **Spatial scope:**
  - - node
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  - **- continent (N.B., XY = country other than DE)**
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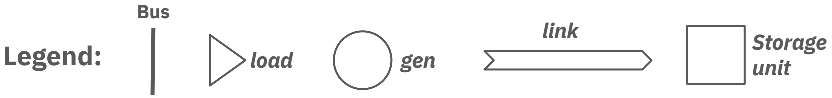


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